

Anti-Jamming Resource Allocation in Air-Terrestrial Integrated Networks: A Hierarchical Game-Theoretic MADRL Approach

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Abstract—This paper investigates the multi-dimensional resource allocation in air-terrestrial integrated networks under jamming environments, where task requirements, the mobility of unmanned aerial vehicles (UAVs), network topology, and anti-jamming spectrum access decisions are strongly coupled. Besides, the heterogeneous attributes of UAVs and ground users (GUs) further complicate the solution of the multi-dimensional optimization problem. Thus, the problem is first formulated as a hierarchical game model comprising two layers. i) a high-layer partially observable stochastic game (POSG) for joint UAV trajectory optimization and GUs channel selection, and ii) a low-layer coalition formation game (CFG) for managing dynamic network associations. It is proved that the low-layer CFG converges to a stable coalition structure using exact potential game (EPG) theory. In addition, we propose a novel multi-agent coalition formation learning framework, and accordingly design a coalition-based heterogeneous multi-agent proximal policy optimization algorithm that dynamically decouples UAV movement, GUs' channel access, and adaptive UAV-GU associations under task-specific demand. Meanwhile, a novel metric named Quality of Task-completion (QoT) is proposed to quantify how well communication systems meet the heterogeneous demands of tasks, and a hierarchical reward mechanism is designed to enhance coordination between UAVs and GUs. Extensive simulation results demonstrate the superiority of our approach over existing benchmarks in QoT, jamming robustness, and energy efficiency.

Index Terms—Heterogeneous multi-agent deep reinforcement learning, air-terrestrial integrated networks, anti-jamming, coalition formation game.

I. INTRODUCTION

IN RECENT years, the low-altitude economy has emerged as a significant frontier, powering applications from logistics and surveillance to emergency communications. A key enabler of this ecosystem is the air-terrestrial integrated communication network [1], which leverages UAV-mounted

platforms to form a promising architecture for flexible coverage extension and capacity augmentation in underserved areas [2]. Owing to their deployment flexibility and superior line-of-sight link probability, such integrated networks have attracted considerable research interest [3]. These systems typically involve two heterogeneous agent types: unmanned aerial vehicles (UAVs) that are dynamically deployed to provide spectrum access services, and ground users (GUs) that communicate with these platforms. This operational paradigm gives rise to two fundamental coupled decision tasks: i) UAV trajectory design [4], where UAVs dynamically reposition to maintain service quality and coverage, and ii) GU spectrum access [5], where GUs must intelligently select transmission parameters like channel and power to mitigate both internal co-channel interference and external malicious jamming [6], [7]. This paper specifically addresses the critical challenge of executing these communication tasks effectively in hostile jamming environments, modeling the UAVs and GUs as a team of heterogeneous agents that must form dynamic, cooperative coalitions for robust jamming defense.

Orchestrating such cooperative anti-jamming strategies is highly challenging. The system is inherently distributed, with UAVs and GUs operating based on local observations. Furthermore, the combined effect of dynamic malicious jammers [8] employing agile frequency hopping and the frequent mobility of UAVs induces rapid and unpredictable fluctuations in the spectrum environment and network topology, significantly escalating the problem's complexity and dynamism. While traditional methodologies, including game theory [9], convex optimization [10], and reinforcement learning (RL), have been widely applied to resource allocation in wireless networks, they often fall short in addressing two critical challenges particular to this setting: 1) The strong coupling among multi-dimensional factors, including dynamic task demands, UAV mobility, time-varying network topology, and distributed spectrum access, which creates a complex, non-convex optimization problem that is intractable for conventional methods. 2) The inherent heterogeneity of UAVs and GUs poses another layer of complexity. While agent heterogeneity is recognized in multi-agent learning [11], [12], the potential performance gains from explicitly leveraging the distinct roles and capabilities of UAVs and GUs in coordinated formations for jamming defense are often overlooked.

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The limitations mentioned above motivate our investigation into a game-theoretic heterogeneous multi-agent deep reinforcement learning (MADRL) approach for task-driven resource allocation in air-terrestrial integrated networks in jamming environments, tackling the coupled challenges of distributed UAV deployment, dynamic channel selection, and adaptive network association. The adoption of game-based MADRL [13] is motivated by two key factors: 1) Game models provide rigorous mathematical tools to analyze strategic interactions among UAVs and GUs under partial observability, distinguishing it from single-agent RL or centralized optimization, and 2) The hybrid MADRL architecture naturally supports heterogeneous action spaces while ensuring convergence through game-theoretic mechanisms [14], [15].

In summary, the key contributions include:

- **A hierarchical game-theoretic anti-jamming model is formulated**, aiming to tackle the strong coupling multi-dimensional resource allocation problem, which decouples the decision-making process by representing joint UAV mobility and GU channel access as a partially observable stochastic game (POSG) at the high layer, and dynamic UAV-GU association management as a coalition formation game (CFG) at the low layer.
- **A multi-agent hierarchical coalition learning framework is proposed**, and designed around a continuous, bidirectional interaction between layers. The high-level POSG controller's decisions define the state space for the low-level CFG optimizer; furthermore, the CFG determines UAV-GU associations and forms coalitions to reshape the environment, subsequently feeding back into the high-level model for the next round.
- **A game-theoretic heterogeneous MADRL algorithm is designed**, which leverages the stable coalition formed by the low-layer to inform the policy learning of UAVs and GUs in the high-layer, enabling more coordinated trajectory planning and channel selection. A new performance metric is introduced as the key to drive the hierarchical reward mechanism, quantitatively measuring how well it meets the heterogeneous task demands.

The subsequent sections are organized as follows: Section II reviews existing multi-agent decision-making research on air-terrestrial integrated communication systems. Section III delineates the system model and presents the problem formulation. Building upon this, Section IV represents the optimization problem as a hierarchical game model and proposes a game-theoretical MADRL approach to address it. The numerical simulations and their analyses are discussed in Section V and Section VI shows the conclusion and future works.

II. RELATED WORK

This section reviews resource allocation in air-terrestrial integrated networks, focusing on game-theoretic and RL approaches.

A. Game-Theoretic Methods

Game theory provides a foundational framework for decentralized decision-making in air-terrestrial integrated networks,

with a broad survey of its applications available in [16]. Building on this groundwork, advanced game-theoretic solutions have been developed to address specific challenges. For instance, Ni et al. [17] proposed an improved potential game approach to maximize cooperative coverage by optimizing UAVs' moving directions. Further extending this line of research, the authors in [18] investigated the joint problem of access control and UAVs' deployment. They formulated the access control as an exact potential game, solved its Nash Equilibrium (NE) via a best-response algorithm, and optimized the UAV-BS deployment via convex optimization to maximize the minimum GUs' throughput. Meanwhile, other studies have addressed more dynamic and complex scenarios. Huang et al. [19] tackled the connectivity guarantee problem in jamming environments by introducing a graph coalition formation game (CFG) that optimizes UAVs' deployment, transmission power, and user association to maximize total network utility. Similarly, for collaborative task allocation in heterogeneous UAV networks, Yan et al. [20] combined semi-centralized clustering with a distributed coalition formation scheme to maximize task utility by optimizing association relationships.

However, traditional game-theoretic algorithms are primarily designed for and perform well in relatively stable environments; their efficacy can be limited in unknown dynamic environments, as they often rely on global state information assumed to be static or slowly varying, thereby overlooking the impact of long-term and rapid state transitions. As environments grow increasingly complex and non-stationary, the need for more adaptive, learning-based solutions becomes paramount.

B. DRL-Based Methods

Intelligent learning algorithms, particularly RL, can overcome the limitations of traditional methods [21], [22] by enabling online, real-time resource allocation through interaction with dynamic and unknown environments. A common approach is to apply single-agent RL algorithms, such as Deep Deterministic Policy Gradient (DDPG), Deep Q-Network (DQN), and Proximal Policy Optimization (PPO), to multi-UAV-assisted air-terrestrial integrated networks. For instance, to tackle the problem of energy-efficient UAV control for fair communication coverage, Liu et al. [23] developed a DDPG-based algorithm which optimizes UAV movement direction to maximize coverage. Furthermore, [24] employed DQN for the joint optimization of UAV trajectory and transmission power, harnessing UAV mobility to maximize real-time downlink capacity within coverage constraints. To minimize the UAV's service time, Zhang et al. [25] introduced a DRL algorithm based on PPO for jointly optimizing a set of flight parameters, namely speed, direction, and altitude, while guaranteeing user QoS. The above approaches have achieved some success, however, straightforwardly extending single-agent reinforcement learning algorithms to multi-agent scenarios poses significant challenges, e.g., non-stationary environmental feedback and inter-agent strategy dependencies.

To mitigate these issues, prior work [11], [26], [27], [28], [29] introduced MADRL methods employing

centralized training with decentralized execution (CTDE) for fair resource allocation, where each UAV learns a policy based on global information during training and executes actions by local observation in the execution. For example, Qin et al. [26] adopted the multi-agent deep deterministic policy gradient (MADDPG) algorithm to optimize UAV trajectories for maximizing fair coverage. In [27], Qian et al. proposed a bi-level optimization approach integrating multi-agent actor-critic (MAAC) and the Lagrangian multiplier method to simultaneously optimize UAV trajectory, transmit power, and computational resource allocation for minimizing system-wide energy consumption. Kang et al. [28] introduced an optimization method based on Multi-Agent Proximal Policy Optimization (MAPPO) to jointly optimize spectrum allocation, user association, and task offloading decisions, aiming to minimize task processing latency. Building on the premise that both UAVs and GUs are potential agents, Ding et al. [29] proposed a joint trajectory design and access control algorithm leveraging MADDPG. Their method employs a multi-objective setup for the agents to optimize for both energy efficiency and fairness. Furthermore, [11] proposed a heterogeneous coordinated QMIX (HC-QMIX) algorithm, which expands QMIX to heterogeneous agent scenarios to optimize UAVs' trajectory, GUs' transmit power level, and association relationships for maximizing throughput.

The aforementioned MADRL methods have demonstrated significant advancements in mitigating the non-stationarity problem in multi-agent environments. Despite their success, their purely data-driven nature often fails to exploit the inherent strategic structure of the problem. In scenarios involving multiple coupled variables (such as trajectory, channel, and association) and mixed cooperative-competitive interactions, the strategy space grows prohibitively large. This leads to slow convergence, unstable training, and poor generalization. These limitations motivate the integration of game-theoretic principles into MADRL [13], combining structural priors with adaptive learning to achieve more efficient and robust multi-agent coordination.

C. Motivation

In real-world settings, two critical challenges further complicate the decision-making process. First, in jamming environments, malicious jammers can release high-power signals to disrupt UAV communications, which may compromise data transmission and potentially induce cascading failures and even collision accidents. Second, the tasks generated by GUs are often time-varying and heterogeneous, requiring GUs to dynamically switch their associations among different types of UAVs to meet diverse task requirements. To address these coupled challenges in a structured manner, we propose a hybrid game-theoretic DRL framework with a dual-layer architecture. In the high layer, a MADRL scheme is employed to optimize UAVs' trajectories and GUs' channel access, enabling adaptive responses to dynamic environments and jamming threats. In the low layer, coalitional game theory is applied to efficiently resolve the discrete association problem among heterogeneous

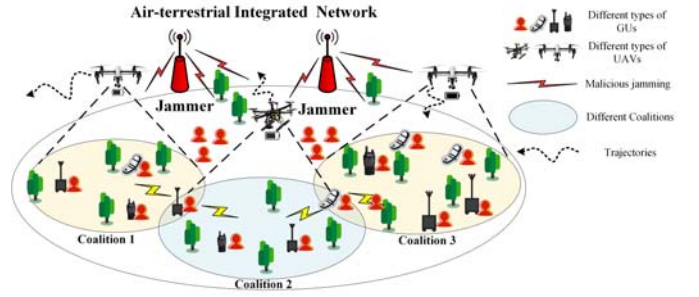


Fig. 1. Air-terrestrial integrated networks in jamming scenarios.

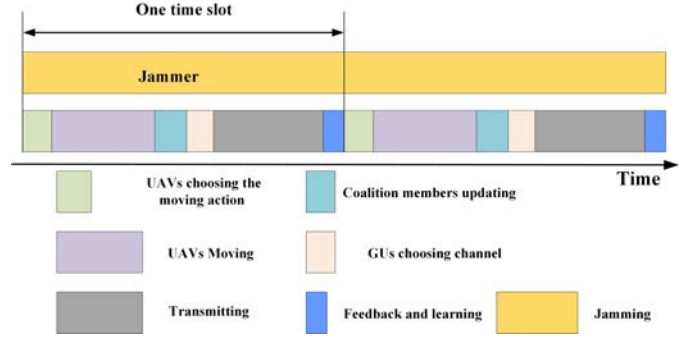


Fig. 2. Time slots structure.

UAVs and GUs, ensuring stable and task-oriented partnerships. This co-design not only enhances learning efficiency by decomposing complex problems into structured levels but also promotes robust, adaptive coordination in complex, adversarial settings.

III. SYSTEM MODEL AND PROBLEM FORMULATION

A. System Model

As depicted in Fig. 1, we consider a downlink air-terrestrial integrated network [30] in which multiple UAVs optimize their trajectories to serve GUs using Orthogonal Frequency-Division Multiple Access (OFDMA). In this scenario, malicious jammers adaptively transmit high-power signals to disrupt the communications. Denoting $\mathcal{N} = \{1, \dots, n, \dots, N\}$ and $\mathcal{M} = \{1, \dots, m, \dots, M\}$ as the set of UAVs and GUs respectively, $\mathcal{T} = \{1, \dots, t, \dots, T\}$ denotes the time slots and its length is represented by τ . At time slot t , assuming that the GU m generates a task $\mathbf{T}_t^m$ with probability P , $\mathbf{T}_t^m$ has its corresponding communication requirements. Its mapping to the communication layer can be simplified to communication rate requirements and coverage duration requirements, denoting $\mathbf{T}_t^m = \{\text{Rate}_t^m, \text{Cover}_t^m\}$. UAVs are managed by various suppliers to provide differentiated services to GUs, thereby alleviating the load on traditional wireless networks [31]. Here, differentiated services refer to offering the same type of service but with varying levels of capability and pricing [32].

Due to the spatial heterogeneity of the spectrum resource, at each time slot t , each GU selects an appropriate transmission channel based on its local spectral environment and reports this decision to the UAVs via a control channel. Upon receiving

the channel selection, the UAV transmits data to the GUs. Each UAV can serve at most k GUs, and a GU can access the network stably only when within the coverage range of a specific UAV. Furthermore, the network is threatened by several malicious jammers that disrupt communications by emitting high-power jamming signals on multiple channels. These jammers operate with fixed transmit power in two patterns: sweep and comb jamming. Consequently, the quality of service for a GU at this time slot depends on the UAV's location, the selected transmission channel, and the association relationship between UAVs and GUs. Thus, as shown in Fig. 2, the time slot structure can be divided into five steps: the UAV selects a movement action, updates the coalition members, and executes the movement action; the GUs complete channel selection based on the local spectrum environment; and finally, the transmission is completed.

B. Mobility Model

The position of GU m at time slot t is given by $q_t^m = [x_t^m, y_t^m, 0]$, with the assumption that all GUs are stationary, but the analysis is also applicable to mobile scenarios. To simplify the model, we assume that all UAVs fly at altitude H with speed V_t^n . Thus, the position of UAV agent n at time slot t is represented by $w_t^n = [x_t^n, y_t^n, H]$. Then the position w_t^n and speed V_t^n of UAV n have the following relationship:

$$\|w_t^n - w_{t-1}^n\| = V_t^n, \quad \forall n \in \mathcal{N}, \quad \forall t \in \mathcal{T} \setminus \{1\}. \quad (1)$$

Moreover, the trajectories of any UAV n and n' are required to conform to predefined collision avoidance constraints, which can be represented as:

$$\|w_t^n - w_t^{n'}\| > d_{\text{safe}}, \quad \forall n \neq n' \in \mathcal{N}, \quad \forall t \in \mathcal{T}, \quad (2)$$

where d_{safe} denotes minimum safe distance between any UAVs. Due to the limited sensing range d_s of UAV n , GU m can only be detected and connected to the network by the UAVs when the distance between them fills $d_t^{m,n} \leq d_s$.

$$\mathbf{A}_t = \begin{bmatrix} a_t^{1,1} & a_t^{1,2} & \cdots & a_t^{1,N} \\ a_t^{2,1} & a_t^{2,2} & \cdots & a_t^{2,N} \\ \vdots & \vdots & \ddots & \vdots \\ a_t^{M,1} & a_t^{M,2} & \cdots & a_t^{M,N} \end{bmatrix}, \quad a_t^{m,n} \in \{0, 1\}. \quad (3)$$

The connection status between UAVs and GUs is represented by a matrix $\mathbf{A}_t$, where the $(m, n)^{\text{th}}$ element, denoted by $a_t^{m,n}$, is a binary variable such that $a_t^{m,n} \in \{0, 1\}$. Here, $a_t^{m,n} = 1$ if and only if GU m is connected to UAV n during time slot t ; otherwise, $a_t^{m,n} = 0$.

C. Communication and Channel Model

Once $a_t^{m,n} = 1$, the communication link between GU m and UAV n can be established. For the communication model and channel model, we consider both transmitter-receiver distance and elevation angle [33] to compute the channel power gain, denote $\mathcal{L} = \{1, \dots, l, \dots, L\}$ as the set of available channels, $L > M$, which implies that each GU can occupy one specific channel for data transmission. Furthermore, multiple adjacent GUs may share the same channel. The channel gain $h_t^{m,n}$

between UAV n and GU m at time slot t can be computed by:

$$h_t^{m,n} = 20 \log_{10}(f_t^m) + 10 n_1 \log_{10}(d_t^{m,n}) + 10 n_2 \log_{10}(e_t^{m,n}) + X_\sigma, \quad (4)$$

where f_t^m is the carrier frequency, $d_t^{m,n}$ and $e_t^{m,n}$ are the respective distance and elevation angle between the UAV n and GU m at time slot t , X_σ denotes the shadow fading, n_1 and n_2 are parameters denoting the elevation angle factor and path loss exponent.

The transmit power is fixed and predefined for each type of serving UAV. Let p_t^n and p_t^j denote the transmit power of UAV n and the jamming power of jammer j at time slot t . The signal-to-interference-plus-noise ratio (SINR) at GU m when receiving from UAV n is expressed as follows:

$$\begin{aligned} \text{SINR}_t^{m,n} &= \frac{p_t^n h_t^{m,n}}{I_t^G + I_t^J + N_0 B} \\ \text{s.t. } p_t^n &\in (0, p^{\max}], \quad \forall n \in \mathcal{N}, \quad t \in \mathcal{T} \\ I_t^G &= \sum_{m \in \mathcal{M}, m \neq m'} \delta(f_t^m, f_t^{m'}) p_t^n h_t^{m',n} \\ I_t^J &= \sum_{j \in \mathcal{J}} \delta(f_t^j, f_t^m) p_t^j h_t^{m,j}, \end{aligned} \quad (5)$$

where $\mathcal{J}$ denotes the set of malicious jammers, $p^{\max}$ denotes the maximum transmit power of UAV, $h_t^{m,j}$ denotes the channel power gain between malicious jammer j and GU m , N_0 represents the power spectral density of additive white Gaussian noise, whereas B denotes the spectral bandwidth, $\delta(f_1, f_2)$ denotes Kronecker delta function, defined as follows:

$$\delta(f_1, f_2) = \begin{cases} 1, & \text{if } f_1 = f_2 \\ 0, & \text{if } f_1 \neq f_2. \end{cases} \quad (6)$$

Once GU m accesses the network, the expected transmission rate $R_t^{m,n}$ is defined as:

$$\begin{aligned} R_t^{m,n} &= \begin{cases} B \cdot \log_2(1 + \text{SINR}_t^{m,n}), & \text{SINR}_t^{m,n} > \text{SINR}^{\text{th}} \\ 0, & \text{SINR}_t^{m,n} < \text{SINR}^{\text{th}} \end{cases}, \end{aligned} \quad (7)$$

where SINR^{th} indicates the received threshold of GU's receiving antenna. From the perspective of task requirement, the current quality of Task-completion (QoT) for $\mathbf{T}_t^m$ served by UAV n at time slot t' can be shown as follow:

$$\text{QoT}_t^{m,n} = \min \left\{ 1, \frac{R_{t'}^{m,n}}{\text{Rate}_t^m} \right\}, \quad (8)$$

where $\text{QoT}_t^{m,n} \in [0, 1]$.

Consequently, the QoT of $\mathbf{T}_t^m$ through Cover $_t^m$ can be written as:

$$\text{QoT}_t^m = \frac{\sum_{t'=t+1}^{t+\text{Cover}_t^m} \sum_{n=1}^N \text{QoT}_{t'}^{m,n}}{\text{Cover}_t^m}, \quad (9)$$

where the numerator term indicates that once the current transmission rate meets the GU's task requirements, further rate increases will no longer enhance the GU's instantaneous QoT in that time slot, which remains contingent upon the anti-jamming performance of this time slot. The denominator term

demonstrates that the GU's QoT is also dependent on the actual transmission duration. Only when both the transmission duration and the actual transmission rate simultaneously satisfy the requirements can the GU m achieve the maximum QoT for $\mathbf{T}_t^m$.

D. Problem Formulation

Owing to the finite energy capacity of the UAVs, energy efficiency needs to be considered. Both flying and transmission will consume energy. To simplify the energy consumption model, we assume that UAV n flies at altitude H with speed V_t^n and expend $\Delta E(V_t^n)$ energy. Our approach works with arbitrary energy consumption models $\Delta E(\bullet)$, provided they satisfy two physical constraints: 1) motion consumes more energy than hovering, and 2) energy expenditure grows monotonically with UAV velocity V_t^n . Namely,

$$\Delta E_t^n = \begin{cases} E_{\text{hover}}, & \text{if } \|V_t^n\| = 0 \\ E_{\text{move}} = W(V_t^n), & \text{others} \end{cases}, \quad (10)$$

where E_{hover} and E_{move} denote moving energy consumption and hover energy consumption, respectively, and $E_{\text{hover}} < E_{\text{move}}$, $W(V_t^n)$ denote the energy consumption of UAV n , it's calculated based on the speed V_t^n .

Our goal is to maximize the total system QoT under limited energy constraints, which necessitates a joint optimization of UAV trajectories, UAV-GU associations, and channel access strategies. To this end, the optimization problem can be mathematically formulated as follows:

$$\mathcal{P} : \max \sum_{t \in \mathcal{T}} \sum_{m \in \mathcal{M}} \frac{QoT_t^m}{\sum_{n \in \mathcal{N}} \Delta E_t^n}. \quad (11)$$

In summary, our goal is to develop a decision-making strategy for UAVs and GUs that maximizes a unified objective function integrating total system QoT and energy efficiency under jamming threats. The formulated problem constitutes a strongly coupled, mixed-integer nonlinear programming problem, which is NP-hard. This inherent complexity, compounded by the need to adapt to dynamic jamming in real time, renders conventional optimization methods computationally intractable. To address this challenge, we reformulate the problem as a hierarchical game model and propose a novel solution that synergizes game theory and MADRL, which is well-suited for finding high-quality policies in such complex and dynamic environments.

IV. GAME-THEORETIC JOINT OPTIMIZATION SCHEME

This section proposes a task-driven Multi-Agent Hierarchical Coalition Learning Framework (MAHCL) for air-terrestrial integrated networks. The MAHCL architecture is first introduced, emphasizing its core design principles to enable decentralized coordination among heterogeneous UAVs and GUs. We then present a hierarchical game-theoretic model to maximize total QoT under energy constraints, incorporating a coalition formation game for nearly optimal UAV-GU associations and a stochastic game for joint mobility control and channel selection. Subsequently, we elaborate on the task-driven distributed UAV-GU coalition formation algorithm,

followed by a detailed implementation of the HAPPO-based moving and channel selection algorithm. Finally, this section analyzes the computational complexity.

A. Multi-Agent Hierarchical Coalition Formation Game Learning Framework

As illustrated in Fig. 3, the proposed task-driven coalition formation game learning framework comprises three key components: network information, an air-terrestrial integrated network, and a task-driven theoretic learning algorithm. The high-level algorithm deployed on the UAVs is responsible for making decisions on movement directions and dynamically forming coalitions with GUs. Specifically, coalitions are constructed by comprehensively considering the location information of both UAVs and GUs, along with the task requirements of GUs. In contrast, the low-level algorithm, implemented at the GUs, focuses on local optimization by selecting the most suitable unoccupied channel to satisfy the QoT requirements of heterogeneous tasks based on the actual established links and real-time interference conditions to complete the communication process.

B. CFG-Based UAV-GU Association Model

Definition 1 (CFG [34]): All GUs are divided into some non-overlapping coalitions, each led by a UAV as the coalition head. Each GU $m \in \mathcal{M}$ has its own coalition preference order $>_m$. The pair $(\mathcal{M}, >_{\mathcal{M}})$ contributes to the proposed CFG. GU m may dynamically transfer between coalitions according to its individual preference $>_m$.

Each GU can choose to leave its current coalition and join a new coalition to enhance QoT (Provided the GU is within range of multiple UAVs). Once GU m join coalition C^n , its coalition formation utility $u^m(C^n, C^{-n})$ is defined as

$$u^m(C^n, C^{-n}) = QoT[C^n] - QoT[C^n \setminus m], \quad (12)$$

where $QoT[C^n]$ denotes the total QoT of C^n , $QoT[C^n/m]$ the sum QoT of C^n except for m , and C^{-n} denotes the coalition selections of other GUs except for GU m (Here, we have decoupled the complex problem. The term QoT below refers to the ideal scenario of QoT). Thus, $u^m(C^n, C^{-n})$ means the increment of total QoT caused by the participation of GU m .

The final coverage performance and coalition partitioning depend on $>_{\mathcal{M}}$. Here, two conventional preference orders are introduced, including selfish and Pareto order.

Definition 2 (Selfish Order [35], [36]): If GU $m \in C^{n'}$ and $\{C^n, C^{n'}\}, \{n, n'\} \in \mathcal{N}$ satisfy the following equation, the selfish order is performed.

$$C^n >_m C^{n'} \Leftrightarrow QoT[C^n \cup m] > QoT[C^{n'} \setminus m], \quad (13)$$

where $C^n >_m C^{n'}$ denotes that GU $m \in C^{n'}$ is inclined to leave the current coalition $C^{n'}$ and join the new coalition C^n .

In the selfish order, each coalition member exclusively focuses on individual utility maximization, prioritizing personal payoff without regard to the collective utility changes within their current coalition [37].

Definition 3 (Stable Coalition Formation [38]): A coalition structure is stable if no GU m can unilaterally change their

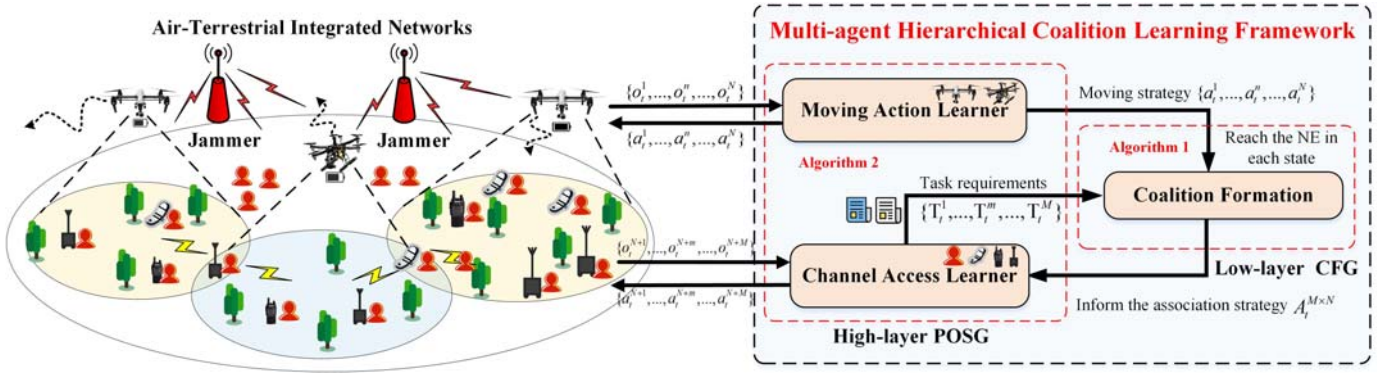


Fig. 3. The Multi-agent hierarchical coalition learning framework.

coalition membership to strictly increase their own utility, which can be written as (14).

$$u^m(C^n, C^{-n}) \geq u^m(C^{n'}, C^{-n'}), \quad \forall m \in \mathcal{M}, \quad \forall C^n, C^{n'} \in \left\{ \bigcup_{n=1}^N C^n \right\}, n \neq n'. \quad (14)$$

Theorem 1: Given the preference relations defined by self-ish order, the proposed UAV-GU association model provably converges to a stable coalition partition while achieving a suboptimal solution.

Definition 4 (Nash Equilibrium [39]): A coalition formation profile $C^* = (C^{1*}, C^{2*}, \dots, C^{N*})$ constitutes a Nash Equilibrium (NE) if and only if formula (15) is satisfied. At the NE, no GU m can unilaterally alter their coalition membership to achieve a strict increase in its utility, given that all other participants keep their coalition choices fixed. It can be formally expressed as:

$$u^m(C^{n*}, C^{-n*}) \geq u^m(C^n, C^{-n}), \quad \forall n \in \mathcal{N}, \quad \forall m \in \mathcal{M}, C^{n*} \neq C^n. \quad (15)$$

Definition 5 (Exact Potential Game [40]): An exact potential game requires the existence of a potential function Φ such that, for any coalition's strategy change, the utility variation equals Φ 's variation. It can be formally expressed as (16):

$$u^m(C^n, C^{-n}) - u^m(C^{n'}, C^{-n'}) = \Phi(C^n, C^{-n}) - \Phi(C^{n'}, C^{-n'}). \quad (16)$$

Theorem 2: The proposed UAV-GU association CFG constitutes an EPG when it serves as its potential function, guaranteeing at least one NE solution.

Proof: The potential function is designed as the sum QoT of all UAV $\mathcal{N}$, which can be written as (17):

$$\Phi(C^n, C^{-n}) = \sum_{n=1}^N QoT[C^n]. \quad (17)$$

The potential function increment $\Delta\Phi = \Phi(C^n, C^{-n}) - \Phi(C^{n'}, C^{-n'})$ resulting from a GU's coalition change from $C^{n'}$ to C^n can be by (18):

$$\begin{aligned} \Delta\Phi &= QoT[C^n \cup m] + QoT[C^{n'} \setminus m] \\ &\quad + QoT[\{C^n\}_{n \in \mathcal{N} \setminus \{C^n \cup m\}} \setminus \{C^{n'} \setminus m\}] - QoT[C^n] \\ &\quad - QoT[C^{n'}] - QoT[\{C^n\}_{n \in \mathcal{N} \setminus \{C^n\}} \setminus \{C^{n'}\}]. \end{aligned} \quad (18)$$

Since the GU's coalition switching only impacts GUs in the current and new coalitions without affecting other coalitions' QoT , we derive (19):

$$\begin{aligned} QoT[\{C^n\}_{n \in \mathcal{N} \setminus \{C^n \cup m\}} \setminus \{C^{n'} \setminus m\}] \\ = QoT[\{C^n\}_{n \in \mathcal{N} \setminus \{C^n\}} \setminus \{C^{n'}\}]. \end{aligned} \quad (19)$$

Thus, the $\Delta\Phi$ can be further written as (20)

$$\begin{aligned} \Delta\Phi &= QoT[C^n \cup m] + QoT[C^{n'} \setminus m] \\ &\quad - QoT[C^{n'}] - QoT[C^n] \\ &= (QoT[C^n \cup m] - QoT[C^n]) \\ &\quad - (QoT[C^{n'}] - QoT[C^{n'} \setminus m]) \\ &= u^m(C^n, C^{-n}) - u^m(C^{n'}, C^{-n'}). \end{aligned} \quad (20)$$

Finally, Theorem 2 is proved, and the proposed UAV-GU association CFG constitutes an EPG with at least one NE point existence.

Algorithm 1 Distributed UAV-GU Coalition Formation Algorithm

Input: The set of GUs $\mathcal{M}$ and UAVs $\mathcal{N}$.

Output: Stable coalition formation $\{C^n\}_{n \in \mathcal{N}}$.

- 1: **Initialize:** Each GU $m \in \mathcal{M}$ randomly selects a UAV $n \in \mathcal{N}$ within its coverage, forming initial coalitions $\{C^n\}$.
- 2: **Repeat:**
- 3: Randomly select a GU m currently in coalition C^n .
- 4: For each UAV $n' \in \mathcal{N} \setminus \{n\}$ within coverage of GU m , compute the utility $u^m(C^{n'}, C^{-n'})$ if m were to join n' .
- 5: Let $n^* = \arg \max_{n'} u^m(C^{n'}, C^{-n'})$, and let u^* be the corresponding utility.
- 6: **if** $u^* > u^m(C^n, C^{-n})$ **then**
- 7: GU m switches from coalition C^n to C^{n^*} :
- 8: Set $a_t^{m,n} \leftarrow 0$ and $a_t^{m,n^*} \leftarrow 1$.
- 9: Update coalitions: $C^n \leftarrow C^n \setminus \{m\}$, $C^{n^*} \leftarrow C^{n^*} \cup \{m\}$.
- 10: **end if**
- 11: **Until:** No GU can improve its utility by unilaterally changing its coalition, meaning a NE is reached.

The proposed coalition formation game between UAVs and GUs constitutes an EPG, where the potential function aligns precisely with the coalition formation utility changes during the coalition updates. Given the finite sets of GUs $\mathcal{M}$

and UAVs $\mathcal{N}$, the space of possible coalition structures is inherently bounded. According to the properties of EPG, this guarantees convergence to an NE coalition partition within finite iterations, where no GU can further improve its utility by leaving its current coalition. Although the solution may not achieve global optimality due to the distributed nature of the process, the algorithm efficiently attains a suboptimal UAV-GU association while ensuring stability. As detailed in Algorithm 1, the iterative switching mechanism progressively refines the coalition structure until equilibrium, leveraging the inherent convergence properties of EPG.

C. POSG-Based Moving and Channel Selection Model

The collaborative decision-making between UAVs and GUs is instrumental in influencing the overall system *QoT* performance. Moreover, each action taken at the current time step drives the state transition from the present to the next state. To capture these dynamics, the movement and channel-selection problems are formulated as a POSG model [41], denoted by $\mathcal{G} = \langle \mathcal{I}, \mathcal{S}, \{\mathcal{A}^i\}_{i \in \mathcal{I}}, P, \{R^i\}_{i \in \mathcal{I}}, \gamma, \mathcal{O} \rangle$, where $\mathcal{I} = \{\mathcal{M} \cup \mathcal{N}\}$. The key elements of $\mathcal{G}$ are as follows:

- $\mathcal{I}$ denotes the set of UAVs and GUs.
- $\mathcal{S}$ denotes the set of environmental state shared by all UAVs and GUs.
- $\mathcal{A}^i$ denotes the set of action of agent i , where $\mathcal{A} = \mathcal{A}^1 \times \dots \times \mathcal{A}^{M+N}$ is the joint action sets of all UAVs and GUs.
- $P: \mathcal{S} \times \mathcal{A} \rightarrow \Delta(\mathcal{S})$: For each time slot $t \in \mathcal{T}$, P denotes the transition probability from current state $s_t \in \mathcal{S}$ to state $s_{t+1} \in \mathcal{S}$ in the next time slot after the joint actions $\mathbf{a}_t \in \mathcal{A}$ are taken.
- $R^i: \mathcal{S} \times \mathcal{A} \times \mathcal{S} \rightarrow \mathcal{R}$: R^i denotes the reward function that returns a scalar value to the UAV or GU i for selecting moving actions or channel selection. The rewards have absolute values uniformly bounded by R^{max} .
- γ denotes the discount factor, where $\gamma \in [0, 1]$. γ is used to quantify the agent's preference for future rewards.
- $\mathcal{O}$ is the joint observation set of all UAVs and GUs, where $\mathcal{O} = \mathcal{O}^1 \times \dots \times \mathcal{O}^{M+N}$, $\mathcal{O}^m$ denotes the observation set of agent m .

Ultimately, the stochastic game serves as a framework to describe the decision-making process of this system: At time slot t , the environment has a state $s_t \in \mathcal{S}$, each UAV and GU i obtains observation information o_t^i based on s_t and executes its action a_t^i , simultaneously with all other agents. The joint actions from all UAVs and GUs make the environment transition into the next state $s_{t+1} \sim P(\cdot | s_t, \mathbf{a}_t)$; then, each UAV and GU receives an immediate reward $R^i(s_t, \mathbf{a}_t, s_{t+1})$. Thus, the expected cumulative discount rewards can be written as (21):

$$V^{\pi^i, \pi^{-i}}(s) = \mathbb{E} \left[\sum_{t \geq 0} \gamma^t R^i(s_t, \mathbf{a}_t, s_{t+1}) \mid a_t^i \sim \pi^i(\cdot | s_t), s_0 \right]. \quad (21)$$

The reward R^i obtained by agent i is determined by both its policy π^i and the joint policy $\pi^{-i} \triangleq \{\pi^j\}_{j \neq i}$ of all other agents in the system.

In the formulated game $\mathcal{G}$, since UAV and GU represent heterogeneous agents with distinct observation and action

spaces. Each agent type seeks a behavioral policy that maximizes its discounted cumulative reward through sequential decision-making. Due to the strategic coupling among agents, where actions and policies interdependently influence system dynamics. Therefore, we attempt to use heterogeneous multi-agent reinforcement learning algorithms to solve this problem.

D. Heterogeneous Multi-Agent Proximal Policy Optimization Algorithm

To address the strategic interactions and partial observability in the stochastic game $\mathcal{G}$, we propose a HAPPO framework. This algorithm adapts the CTDE paradigm to accommodate the distinct observation-action spaces of UAVs and GUs while ensuring policy updates remain stable through proximal optimization. Below, we formalize the key components of the algorithm:

- **Observation:** Owing to partial observability and limited sensor range, each UAV agent n obtains an observation o_t^n consisting of its ID n , current coordinate w_t^n , remaining energy level e_t^n and network access indicators $a_t^{m,n}$ for GUs within its coverage area. It can be formally expressed as:

$$o_t^n = \{n, w_t^n, e_t^n, a_t^{n,1}, \dots, a_t^{n,M}\}. \quad (22)$$

For each GU agent $m \in \mathcal{M}$, its observation comprises its ID m , channel selection f_t^m , the channel under jamming f_t^J and network access status indicators $a_t^{m,n}$, then, the observation of GU agent m can be formally defined as:

$$o_t^m = \{m, f_t^m, f_t^J, a_t^{1,m}, \dots, a_t^{N,m}\}. \quad (23)$$

- **Actions:** The UAVs' operational objective is to optimize service coverage for maximizing user connectivity satisfaction. Thus, at time slot t , the action of UAV agent n can be shown as:

$$a_t^n = \{\varphi_t^n\}, \quad (24)$$

where φ_t^n denotes the fly direction of the UAV agent n . Within the MADRL framework, we employ action space discretization to overcome the computational challenges of continuous control optimization. Specifically, the UAV agents' flight directions are mapped to a finite set: $\{left, right, front, back, hover\}$. For GU agents m , its task involves avoiding spectrum conflict between other GUs and defending malicious jamming attacks. Besides, the action of GU can be depicted as:

$$a_t^m = \{f_t^m\}, \quad (25)$$

where $f_t^m \in \mathcal{L}$ represent the channel selection by GU agent m at time slot t .

- **Reward:** The primary objective of this work is the maximization of the system *QoT* throughout the operational period. Therefore, the reward should be designed such that its value is positively correlated with the objective function $\mathcal{P}$. Additionally, it is essential to consider the task completion status, user access association $a_t^{m,n}$, energy efficiency, and the collision constraint R_t^{col} , imposing penalties on agents that violate these constraints. To effectively optimize the total system *QoT*, we separately

design rewards for the UAV and GU. The reward functions can thus be formulated as follows:

$$\begin{aligned} r_t^n &= \sum_{m \in \mathcal{M}} \frac{QoT_t^{m,n}}{\Delta e_t^n} - R_t^{col}, \\ r_t^m &= \sum_{n \in \mathcal{N}} QoT_t^{m,n}, \end{aligned} \quad (26)$$

where R_t^{col} means collision penalty and Δe_t^n denotes normalized energy expended by the UAV during this movement.

- **Learning rule:** Our proposed algorithm adopts the HAPPO framework to coordinate UAV and GU agents in a decentralized manner. As shown in Algorithm 2, the learning rule consists of four key phases: 1) trajectory collection, 2) advantage estimation, 3) sequential policy updates, and 4) critic network optimization. Below, we dissect each component and its implementation details.

- (1). Trajectory collection: First, each GU and UAV i_m selects actions $a_t^{i_m}$ based on the current policy $\pi_{\theta_k^{i_m}}^{i_m}$ where $\theta_k^{i_m}$ denotes the actor network of i_m at k episode, and generates trajectory data $\{(o_t^{i_m}, a_t^{i_m}, R_t^{i_m}, o_{t+1}^{i_m}), \forall i_m \in \mathcal{I}, t \in T\}$, which is then stored in the experience replay buffer $\mathcal{D}$.
- (2). Advantage estimation: Next, batch data is sampled from the replay buffer $\mathcal{D}$, and the joint advantage function $\hat{A}(s, \mathbf{a})$ is calculated through generalized advantage estimation (GAE). Here, $\hat{A}(s, \mathbf{a})$ quantifies the efficiency of joint actions while reducing variance in policy updates. The GAE can be formally expressed as (27).

$$\hat{A}_{\pi}(s, \mathbf{a}) \triangleq Q_{\pi}(s, \mathbf{a}) - V_{\pi}(s), \quad (27)$$

where $Q_{\pi}(s, \mathbf{a})$ and $V_{\pi}(s)$ denote state-action value function and state value function respectively.

- (3). Sequential Policy Update: [42] The agent order is randomly permuted to mitigate update bias. Subsequently, agent policies are updated sequentially by employing importance sampling to measure the action selection difference between new and old policies and limiting the update step size with a clipping mechanism to prevent abrupt policy changes. The PPO-Clip objective function is defined formula as (28), shown at the bottom of the page, and $M^{i_{1:m}}(s, \mathbf{a})$ can be expressed as (29):

$$M^{i_{1:m}}(s, \mathbf{a}) = \left(\prod_{j=1}^{m-1} \frac{\pi_{\theta_{k+1}^{i_j}}^{i_j}(a^{i_j} | s)}{\pi_{\theta_k^{i_j}}^{i_j}(a^{i_j} | s)} \right) \hat{A}(s, \mathbf{a}), \quad (29)$$

where $M^{i_{1:m}}(s, \mathbf{a})$ denotes the dynamic correction term. It is used to offset the impact of policy changes from previously updated agents on the

gradient estimation of the current agent during sequential updates, ensuring that the direction of policy improvement remains correct. Then, $M^{i_{1:m+1}}$ can be written as (30):

$$M^{i_{1:m+1}}(s, \mathbf{a}) = \frac{\pi_{\theta_{k+1}^{i_{m+1}}}^{i_{m+1}}(a^{i_{m+1}} | o^{i_{m+1}})}{\pi_{\theta_k^{i_{m+1}}}^{i_{m+1}}(a^{i_{m+1}} | o^{i_{m+1}})} M^{i_{1:m}}(s, \mathbf{a}). \quad (30)$$

Algorithm 2 HAPPO-Based Moving and Channel Selection Algorithm

Input: Learning rate α , batch size $\mathcal{B}$, replay buffer $\mathcal{D}$, training episodes K , agents $\mathcal{I} = \mathcal{M} \cup \mathcal{N}$, steps per episode T .

Output: Trained policy parameters $\{\theta^i\}_{i \in \mathcal{I}}$ for all agents

- 1: Initialize actor networks $\{\theta_0^i\}_{i \in \mathcal{I}}$, critic network ϕ_0 , replay buffer $\mathcal{D}$.
 - 2: **for** episode $k = 1$ to K **do**
 - 3: Reset environment to initial state s_0 .
 - 4: **{Phase 1: Data Collection}**
 - 5: **for** step $t = 0$ to $T - 1$ **do**
 - 6: **for** each agent $i \in \mathcal{I}$ **do**
 - 7: Observe o_t^i from environment.
 - 8: Sample action $a_t^i \sim \pi_{\theta_k^i}(\cdot | o_t^i)$.
 - 9: **end for**
 - 10: Execute joint action $\mathbf{a}_t = (a_t^1, \dots, a_t^{|\mathcal{I}|})$.
 - 11: Observe reward $\mathbf{r}_t$ and next observations $\mathbf{o}_{t+1}$.
 - 12: Store transition $(\mathbf{o}_t, \mathbf{a}_t, \mathbf{r}_t, \mathbf{o}_{t+1})$ in $\mathcal{D}$.
 - 13: **end for**
 - 14: **{Phase 2: Advantage Estimation}**
 - 15: Sample minibatch of $\mathcal{B}$ transitions from $\mathcal{D}$.
 - 16: Compute advantage estimates $\hat{A}(s, \mathbf{a})$ using GAE.
 - 17: **{Phase 3: Sequential Policy Update}**
 - 18: Randomly permute agent order: $i_1, i_2, \dots, i_{|\mathcal{I}|}$.
 - 19: Initialize $M^{i_1}(s, \mathbf{a}) = \hat{A}(s, \mathbf{a})$.
 - 20: **for** $m = 1$ to $|\mathcal{I}|$ **do**
 - 21: Update agent i_m 's policy using PPO-Clip objective: (28).
 - 22: Compute correction term for next agent: (30).
 - 23: **end for**
 - 24: **{Phase 4: Critic Update}**
 - 25: Update critic network by minimizing MSE loss (31)
 - 26: **end for**
-

- (4). Critic network optimization: Updating global V-value network by formula (31):

$$\phi_{k+1} = \arg \min_{\phi} \frac{1}{\mathcal{B}T} \sum_{b=1}^{\mathcal{B}} \sum_{t=0}^T (V_{\phi}(s_t) - \hat{R}_t)^2, \quad (31)$$

where $\hat{R}_t$ denotes the estimated reward value of state s_t , $\mathcal{B}$ denotes the batch size and T denotes the steps per episode, the objective is to minimize the mean

$$\mathcal{L}^{i_m}(\theta_{k+1}^{i_m}) = \mathbb{E} \left[\min \left(\frac{\pi_{\theta_{k+1}^{i_m}}^{i_m}(a_t^{i_m} | o_t^{i_m})}{\pi_{\theta_k^{i_m}}^{i_m}(a_t^{i_m} | o_t^{i_m})} M^{i_{1:m}}(s_t, \mathbf{a}_t), \text{clip} \left(\frac{\pi_{\theta_{k+1}^{i_m}}^{i_m}(a_t^{i_m} | o_t^{i_m})}{\pi_{\theta_k^{i_m}}^{i_m}(a_t^{i_m} | o_t^{i_m})}, 1 \pm \epsilon \right) M^{i_{1:m}}(s_t, \mathbf{a}_t) \right) \right]. \quad (28)$$

TABLE I
NOTATION AND DEFINITIONS

Notation	Definition
$\mathcal{M}$	Index set $\{1, \dots, m, \dots, M\}$ of UAVs
$\mathcal{N}$	Index set $\{1, \dots, n, \dots, N\}$ of GUs
$\mathcal{J}$	Index set $\{1, \dots, j, \dots, J\}$ of jammers
$\mathcal{T}$	Index set $\{1, \dots, t, \dots, T\}$ of time slots
$\mathcal{L}$	Index set $\{1, \dots, l, \dots, L\}$ of available channels
w_t^n	The position of UAV n
q_t^m	The position of GU m
p_t^n	The transmit power of UAV n
p_t^j	Jamming power from jammer j
f_t^m	Selected channel of GU m
$\mathbf{A}_t$	The association matrix between $\mathcal{M}$ and $\mathcal{N}$
$h_t^{m,n}$	The channel gain between UAV n and GU m
$h_t^{m,j}$	The channel gain between GU m and jammer j
I_t^G	Co-channel interference power from other UAVs
I_t^J	Co-channel jamming power from jammers
$d_t^{m,n}$	The distance between UAV n and GU m
$\text{SINR}_t^{m,n}$	The SINR between UAV n and GU m
$R_t^{m,n}$	The expected transmission between UAV n and GU m
$QoT_t^{m,n}$	The QoT between UAV n and GU m

squared error, thereby making the value estimate $V_\phi(s_t)$ as close as possible to the reward $\hat{R}_t$.

E. Complexity Analysis

The neural architecture comprises distinct modules: $N + M$ actor networks and critic networks. Both networks feature an identical layered structure, including three hidden layers sandwiched between input and output layers. Though their respective functions differ significantly. Let K , T , $\mathcal{B}$, h_a^i and h_c^i denote the number of training episodes, the training steps per episode, the size of mini-batch, the neuron numbers of actor-network and critic-network for i -th layer, respectively. Then, the computational complexity for training $M + N$ agents in the HAPPO is given by $O((M + N)KTB[(\sum_{i=0}^3 h_a^i h_a^{i+1} + \sum_{i=0}^3 h_c^i h_c^{i+1})])$ [43]. Moreover, the computational complexity of the coalition formation algorithm is given by $O(M \times N)$.

V. SIMULATION RESULT

This section provides a performance evaluation of our proposed algorithm for an air-terrestrial integrated communication scenario. First, we detail the simulation parameters, including UAV configurations, channel conditions, and task generation models. We then compare our approach with three benchmark methods: MADQN [44], MADDPG [26], and MAPPO [28]. Finally, we analyze key performance metrics such as QoT and trajectory design under dynamic network conditions.

Consider a downlink communication employing two heterogeneous UAV types [32] operating at constant altitude $H = 100$ m to service GUs within a 1000×1000 m² area under jamming scenarios. These include a sweep jammer and a fixed jammer. Both jammers operate at 40 dBm transmission power. At the end of each time slot, each GU sends an ACK/NACK on a low-rate control channel [12], [45]. The UAVs use this feedback to compute their rewards and maintain temporal and spatial synchronization.

Each UAV features 150 m coverage radius and 250 m sensing range for GUs localization. At time slot t , assumes

TABLE II
SIMULATION PARAMETERS SETTING

Parameters	Descriptions	Values
H	Altitude of UAVs above ground	100 m
B	Available channel bandwidth	1 MHz
σ^2	Noise power spectrum density	-169 dBm
p_t^n	Transmit power of UAV n	9 dBm to 15 dBm
d_c	Coverage radius of UAVs	150 m
d_{safe}	Collision avoidance distance	20 m
V_t	Speed of UAVs	10 m/s
P^{task}	Task generation probability of GU	0.8
Rate_t^m	Rate requirement of GU m	1 kbps to 1.5 kbps
Cover_t^m	Coverage requirement of GU m	3 s to 5 s
p_t^j	Transmit power of jammer j	50 W
E	Total energy budget of UAVs	500 units
E_{hover}	Hovering power consumption	1 unit
E_{move}	Moving power consumption	1.5 units

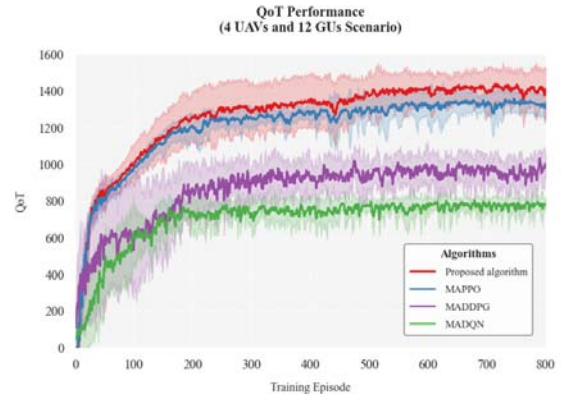


Fig. 4. Total QoT performance over training for different algorithms.

that the GU m generates task Task_t^m with probability P^{task} (provided that the GU m is idle), Task_t^m has its corresponding communication requirements which can be simplified as: i) data rate demand Rate_t^m and ii) coverage duration demand Cover_t^m . Only if located within the coverage area of a specific UAV can a GU request access to the network. The transmit powers of UAVs are 10 mW and 20 mW, respectively. Other parameters are set as follows: bandwidth $B = 1$ MHz, noise PSD $\sigma^2 = -169$ dBm/Hz, UAV speed $V_t = 10$ m/s, the maximum service capacity of UAV is 3 and 4, the UAV protection distance $d_{\text{safe}} = 20$ m. Table II shows the main simulation parameters [46]. Each simulation was repeated 10 times for statistical reliability, with each run consisting of 800 episodes (500 time slots). The learning rate and random seed were $1e-5$ and 1. The benchmarking platform was configured with the Ubuntu 22.04 operating system, an i9-13900K CPU, and an RTX 4080 Ti GPU.

In Fig. 4, we compared our proposed algorithm with MADQN, MADDPG, and MAPPO using 12 GU and 4 UAV agents. The number of available channels is set to 13, and the learning rate is set to $1e-5$. Each episode consists of 500 time slots. The superior performance of our proposed HAPPO with the CFG algorithm, as evidenced in Fig. 4, can be attributed to several key design factors. Firstly, the explicit modeling of agent heterogeneity enables tailored policy learning for UAVs and GUs, addressing the

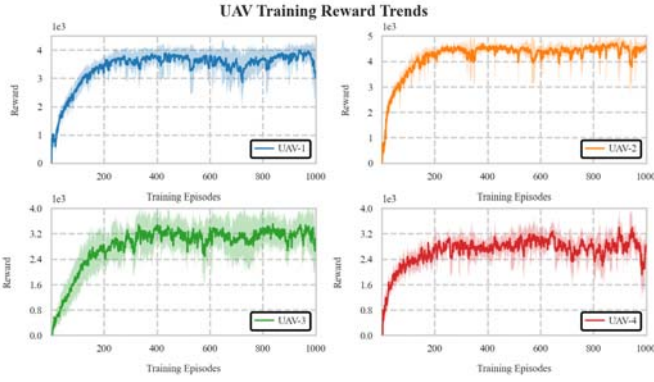


Fig. 5. Reward during training for different UAV.

fundamental limitations of homogeneous algorithms such as MADDPG and MAPPO, which struggle with mismatched action and observation spaces. Secondly, the integration of the CFG provides a stable and theoretically-grounded mechanism for dynamic UAV-GU associations. This directly mitigates the miscoordination in network access that plagues fully decentralized methods like MADQN and reduces the non-stationarity challenge for the learning agents. Thirdly, the sequential policy update scheme in HAPPO, coupled with the hierarchical reward structure, ensures stable and coordinated gradient updates across heterogeneous agents, preventing the performance degradation observed in MAPPO and MADDPG due to simultaneous, conflicting policy updates.

Fig. 5 illustrates the reward trajectories of UAVs during the training process. In the initial phase, all UAVs exhibit rapid performance gains as their coverage of GUs expands. As training progresses and UAVs gradually achieve full coverage of all users, the rate of reward increase slows. This phenomenon arises because the reward mechanism of UAVs jointly depends on the dynamic channel selection strategies and association relationships of GUs. As the GUs channel decision policies converge, the system eventually stabilizes at a steady-state reward equilibrium. Notably, the converged performance ceilings vary across UAVs, which can be attributed to the heterogeneous nature of UAV platforms that directly influence resource allocation efficiency.

As shown in Fig. 6, the UAV agents have learned to form task coalitions with GUs by dynamically optimizing their trajectories and service associations through coalitional game learning. The paths of UAVs and the locations of GUs are represented by colored lines and red dots, respectively. Notably, the UAVs do not simply hover near fixed users; instead, they continuously coordinate their movements and associations based on the task-oriented CFG. The connecting lines between UAVs and GUs, labeled *Co1* to *Co4*, depict real-time coalition structures that adapt to UAV mobility, shifting task demands, and network constraints. This demonstrates the self-organizing capability of the proposed algorithm under dynamic task conditions.

We also compared the performance of the proposed algorithm without using the CFG module, and the benchmark algorithm removes the CFG module, as shown in Fig. 7.

Trajectories of UAVs under dynamic coalitions

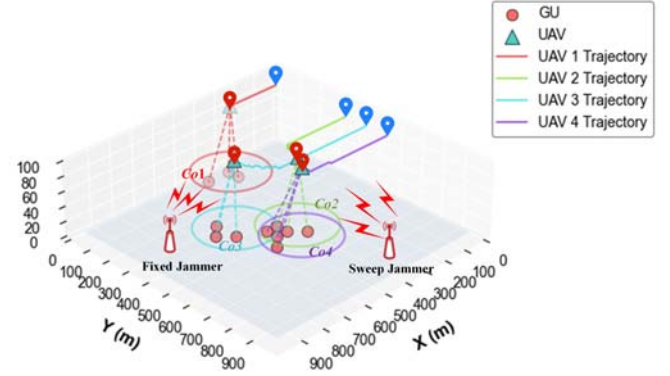


Fig. 6. The process of forming dynamic coalitions.

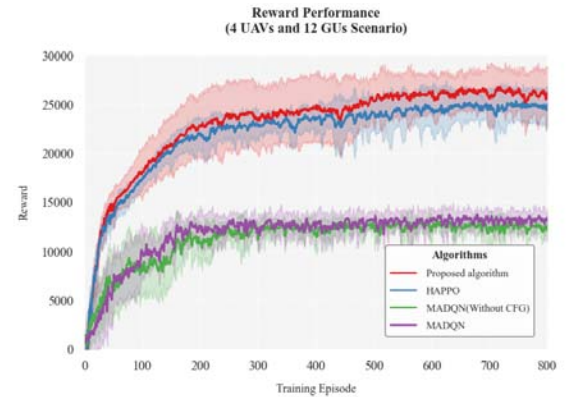


Fig. 7. Reward during training for different algorithms.

In contrast to the benchmark algorithm, where UAVs establish connections by simply linking to their nearest user without coalition formation, the proposed algorithm demonstrates superior performance due to the inherent heterogeneity between GUs' tasks and UAV capabilities. The coalition formation preference enables the construction of stable coalitions tailored to task requirements. Although the resulting coalitions represent suboptimal solutions, they yield enhanced utility gains for both GUs and UAVs compared to proximity-based connection strategies. This performance improvement stems from the algorithm's capacity to coordinate resource allocation while respecting individual rationality constraints during the coalition formation process. The convergence performance of MADQN and MADQN (Without CFG) is not significantly different. This is because the inherently distributed characteristics of these algorithms hinder their ability to find a high-quality solution for UAV trajectory planning.

As shown in Fig. 8, the proposed algorithm demonstrates remarkable robustness across all task probability regimes ($P^{\text{task}} \in 0.2, 0.5, 1.0$). It consistently outperforms the baseline methods, achieving an average performance improvement of approximately 12%. Notably, while the performance of baseline algorithms fluctuates significantly with varying P^{task} , our method maintains a stable and high level of efficacy. This

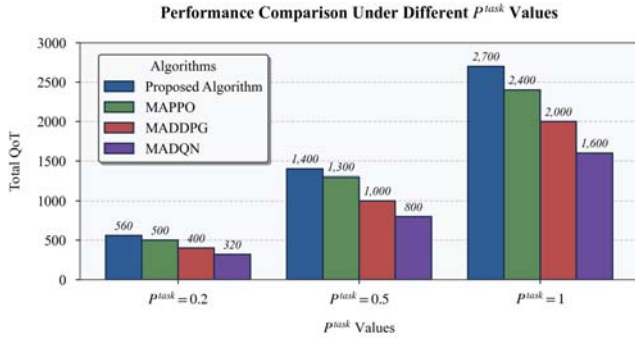


Fig. 8. Algorithm performances under varying task probabilities p^{task} .

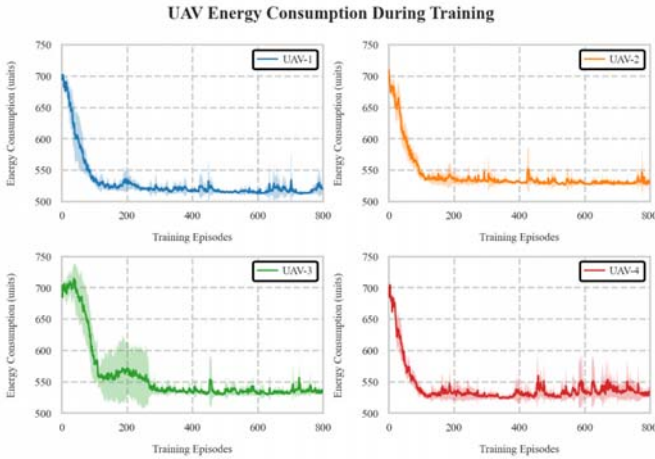


Fig. 9. Energy consumption during training for different algorithms.

stability underscores its adaptability to dynamic task demands, which is a critical requirement in real-world applications.

Fig. 9 presents the convergence characteristics of energy consumption trajectories for UAVs during cooperative task execution. The four subplots demonstrate asynchronous convergence levels of energy consumption across distinct UAV agents. The trajectories converge within 200-300 training episodes, with diminishing variance bounds indicating progressive policy stabilization. Notably, the convergence performance has persistent minor fluctuations, particularly in high user-density regions. These oscillations reflect continuous micro-adjustments in flight trajectories as UAVs dynamically balance energy conservation against coverage requirements in areas with stochastic user distribution patterns.

Furthermore, we conducted runtime evaluations of the proposed approach. The training of 800 episodes (500 steps each) was completed with an average of 3.68 seconds per episode (7.4 ms/step). This per-step latency is well below the 20 ms coherence time typical of Air-Terrestrial integrated networks and also meets the microsecond inference requirements for edge devices such as the Jetson TX2.

VI. CONCLUSION

This paper investigated the problem of maximizing the sum service quality of tasks in task-driven air-terrestrial integrated networks under jamming by jointly optimizing UAV

trajectories, association relationships, and GUs' channel selection. A hierarchical game model based on CFG and POSG is used to analyze the complex interaction between UAVs and GUs. We subsequently proposed a multi-agent coalition formation learning algorithm to attain a near-optimal solution, which is demonstrated through simulations to yield performance that surpasses all benchmark approaches.

Building on this foundation, a natural extension is to investigate a more agile and resilient architecture. Therefore, a promising yet challenging direction is to evolve the current framework towards a fully distributed, control-channel-free system. The primary hurdles include achieving robust distributed consensus on coalition structures and policies without dedicated signaling, mitigating the exacerbated non-stationarity in the learning environment, and designing lightweight communication protocols to embed coordination into data links with minimal overhead. To address these, we will explore theoretical frameworks such as communication-minimizing MADRL and distributed game-theoretic consensus algorithms, while concurrently validating these solutions on integrated SDR-AI-UAV platforms (such as LimeSDR USB, Jetson TX2 [47], and DJI M300 for a 3kg payload) to bridge theory and practice. By combining open-source emulators [48] such as NS-3, the pre-training can be completed first.

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